

3rd Sem.-B.Tech.

END-SEMESTER EXAMINATION, NOV.-DEC.,2025

Course Code- EPMTG301

Course Title- Mathematics for Machine Learning

Time- 03 Hours

Max. Marks-50

Note: - Attempt all five questions. Missing data/ Information if any, may be suitably assumed & mentioned in the answer.

Q.No.	Attempt any 2 parts of the questions below	Marks	CO
1(a)	(i) Check whether the set $W = \{(u, v, w) \in \mathbb{R}^3 : u + 2v - w = 0, 2x - y + z = 0\}$ is a subspace of $\mathbb{R}^3$ with standard addition and multiplication. (ii) If $(\alpha_1, \alpha_2, \alpha_3)$ be a basis of real vector space V and $\beta_1 = \alpha_1 + \alpha_3$, $\beta_2 = 2\alpha_1 + 3\alpha_2 + 4\alpha_3$, $\beta_3 = \alpha_1 + 2\alpha_2 + 3\alpha_3$, check whether $(\beta_1, \beta_2, \beta_3)$ is a basis of V .	05	CO1
1(b)	Let $(x, y, z), (r, s)$ be ordered bases of the real vector spaces V and W respectively and a linear mapping $T: V \rightarrow W$ maps the basis vectors as $T(x) = r + s, T(y) = 2r - s, T(z) = r + 3s$. Find the matrix of T relative to the ordered bases (i) (y, z, x) of V and (r, s) of W (ii) $(x + y, y, z)$ of V and $(r, r + s)$ of W .	05	CO1
1(c)	Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the transformation given $T(x, y, z) = (x, 0, x - y + z)$. Find a basis for $\text{Ker}T$ and $\text{Range}T$. Also verify Rank-Nullity Theorem.	05	CO1
2(a)	Find the value of θ for which the quadratic form $q(x, y, z) = x^2 + \theta y^2 + 2yz + \theta z^2$ is positive definite. Also find the Eigen Values of the associated matrix at $\theta = 2$.	05	CO2
2(b)	Use Gram-Schmidt process to obtain an orthonormal basis of subspace of $\mathbb{R}^4$ with the inner product defined by $\langle \alpha, \beta \rangle = a_1b_1 + a_2b_2 + a_3b_3 + a_4b_4$ where $\alpha = (a_1, a_2, a_3, a_4), \beta = (b_1, b_2, b_3, b_4)$, generated by the basis $\{(1, 0, 1, 0), (1, 1, 1, 1), (0, 1, 2, 1)\}$	05	CO2
2(c)	(i) Let $u = (1, 1, 1), v = (1, 2, 3), w = (1, 3, 4)$. Find $r = (a, b, c)$ such that $\langle u, r \rangle = 10, \langle v, r \rangle = 20, \langle w, r \rangle = 27$, where the inner product defined by $\langle x, y \rangle = 2x_1y_1 + x_2y_2 + x_3y_3$ where $x = (x_1, x_2, x_3), y = (y_1, y_2, y_3)$ (ii) If $u = (-2, -3, 5), v = (6, 9, -15) \in \mathbb{R}^3$. Verify that $ \langle u, v \rangle = \ u\ \ v\ $	05	CO2
3(a)	Find the singular value decompositions of the matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$	05	CO3
3(b)	Find the polar decomposition of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ -2 & 1 & 0 \end{bmatrix}$	05	CO3
3(c)	Find Pseudo pseudo-inverse of the matrix $A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \\ 1 & -1 \end{bmatrix}$	05	CO3

4(a)	If the field $\vec{V} = (y^2 + 2cxz)\hat{i} + y(bx + cz)\hat{j} + (y^2 + cx^2)\hat{k}$ is conservative, find values of b, c . Also, find the work done in moving an object in this field from $(1, -2, 1)$ to $(3, 1, 4)$.	05	CO4
4(b)	Evaluate $\iint_S \vec{A} \cdot \hat{n} dS$ where $\vec{A} = (x + y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}$ and S is the surface of the plane $2x + y + 2z = 6$ in the first octant.	05	CO4
4(c)	If $r^2 = x^2 + y^2 + z^2$, show that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$ and hence deduce that $\nabla^4 f(r) = f^{(4)}(r) + \frac{4}{r} f'''(r)$.	05	CO4
5(a)	Minimize: $f(x, y) = 8x + 10y - x^2 - y^2$ Subject to: $3x + 20y \leq 6$ $x, y \geq 0$ using KKT conditions find the optimal solution.	05	CO5
5(b)	Maximize: $(x + 1)(y + 1)(z + 1)$ Subject to: $x^m y^n z^p = k$, where a, b, c, k are constant.	05	CO5
5(c)	Consider the problem of minimization of $f(x, y) = 4x^2 - 8xy + 6y^2$, over $(x, y) \in \mathbb{R}^2$. Perform 3 iterations of the steepest descent method taking the starting point with $x_0 = (1, 1)^T$	05	CO5